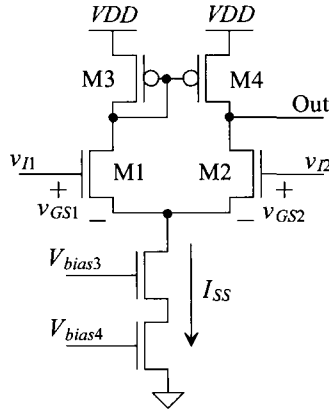




**Figure 22.6** Diff-amp with a current mirror load.

ground. The minimum input common mode voltage is, once again, given by Eq. 22.11. The maximum input common mode voltage is determined knowing that the drain voltage of M2 is the same as the drain voltage of M1 (when both diff-amp inputs are the same potential), that is,  $V_{DD} - V_{SG}$  of the PMOS. We can therefore write

$$V_{DS} \geq V_{GS} - V_{THN} \rightarrow V_D \geq V_G - V_{THN} \rightarrow V_{CMMAX} = V_{DD} - V_{SG} + V_{THN} \quad (22.12)$$

For the parameters in Table 9.1,  $V_{CMMAX} = 5 - 1.15 + 0.8 = 4.65 \text{ V}$ .

The voltage output swing can be determined by noting that the maximum voltage output is limited by keeping M4 in saturation. Therefore,

$$V_{OUTMAX} = V_{DD} - V_{SD,sat} \quad (22.13)$$

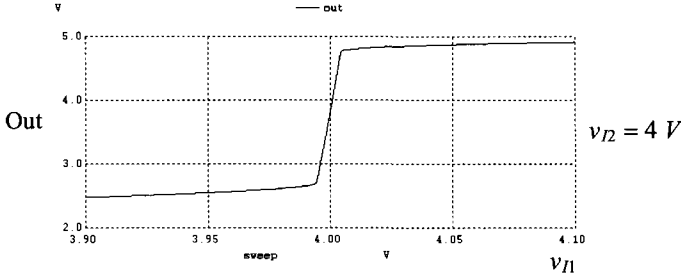
The minimum output voltage is determined by the voltage on the gate of M2 (M2 must remain in saturation).

$$V_D \geq V_G - V_{THN} \rightarrow V_{OUTMIN} = V_{I2} - V_{THN} \quad (22.14)$$

### Example 22.3

Suppose the diff-amp in Fig. 22.6 is implemented with the sizes and bias currents seen in Table 9.1 ( $I_{SS} = 40 \mu\text{A}$ ). If the gate of M2 is held at 4 V, estimate the diff-amp's output swing. Verify the answers with SPICE.

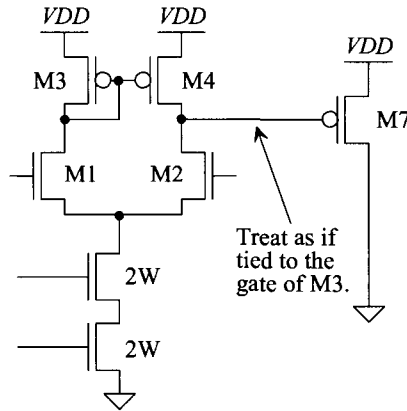
From Eq. (22.13), the output can swing up to 4.75 V before M4 triodes. From Eq. (22.14), the output can swing down to  $4 - 0.8$  or 3.2 V before M2 triodes. The simulation results are seen in Fig. 22.7. Looking at the results, we see the output goes down to approximately 2.8 V or 400 mV less than what we predicted. This can be attributed to the body effect in M1 and M2. With body effect, the threshold voltage is 1.2 V (instead of 0.8 V). ■



**Figure 22.7** Simulation results for Ex. 22.3

### Biasing from the Current Mirror Load

Consider the connection of the common-source amplifier, M7, to the output of the diff-amp in Fig. 22.8. When the inputs to the diff-amp are at the same potential, the currents that flow in M3 and M4 are equal ( $= I_{SS}/2$ ). We know from Ch. 20 that the drain of M4 is then at the same potential as its gate. This means, for biasing purposes, that the gate of M7 can be treated as if it were tied to the gate of M3 (M4).



**Figure 22.8** Using a diff-amp to bias the next stage amplifier.

### Minimum Power Supply Voltage

Looking at the diff-amp in Fig. 22.6, we can estimate the minimum allowable power supply voltage,  $VDD_{min}$ , as

$$VDD_{min} = V_{SG3} + \overbrace{V_{DS,sat1}}^{\text{minimum voltage across M1}} + \overbrace{2V_{DS,sat}}^{\text{minimum voltage across } I_{SS}} \quad (22.15)$$

Looking at this equation, we see that the  $V_{SG}$  of M3 is the “weak link” in the minimum power supply voltage. Using the parameters from Table 9.2 (the short-channel devices with a  $VDD$  of 1 V), we get a  $VDD_{min}$  of 500 mV. We can lower this value by using a single MOSFET to bias the diff-amp (instead of a cascode circuit).

### 22.1.2 AC Operation

To describe the AC small-signal operation of the diff-amp in Fig. 22.1, consider the AC schematic seen in Fig. 22.9. We've replaced the current source with an open and the DC supply ( $VDD$ ) with a short (to ground). In the following discussion, it's important to remember that a negative AC current simply means that the overall current (AC + DC) is decreasing (see Ex. 9.5 in Ch. 9). We can write

$$v_{di} = v_{i1} - v_{i2} = v_{gs1} - v_{gs2} = \frac{i_{d1}}{g_m} - \frac{i_{d2}}{g_m} \quad (22.16)$$

Reviewing Fig. 22.9, we see that

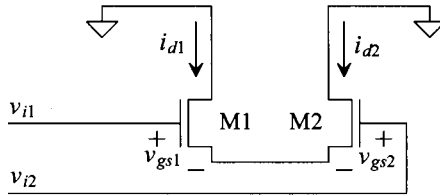
$$i_{d1} = -i_{d2} = i_d \quad (22.17)$$

and thus

$$v_{gs1} = -v_{gs2} = \frac{v_{di}}{2} \quad (22.18)$$

so finally

$$g_m v_{gs1} = g_m (-v_{gs2}) = i_d = g_m \cdot \frac{v_{di}}{2} \quad (22.19)$$



**Figure 22.9** AC circuit for a diff-amp.

#### Example 22.4

Estimate the AC components of the drain currents of M1 and M2 for the circuit in Fig. 22.2 if  $v_{i1} = 2.5 + 1\text{mV} \cdot \sin(2\pi \cdot 1\text{kHz} \cdot t)$ . Verify your hand calculations using SPICE.

From Table 9.1 we know the  $g_m$  of the NMOS devices used in the diff-pair is  $150 \mu\text{A/V}$ . The AC component of  $v_{i2}$  is zero so we know

$$v_{gs1} = -v_{gs2} = 0.5\text{mV}$$

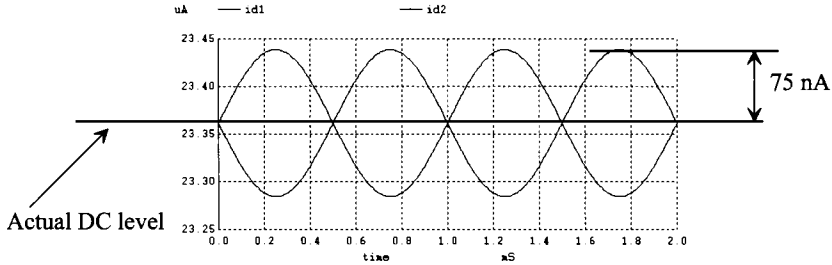
and thus

$$i_d = g_m v_{gs} = (150 \times 10^{-6})(500 \times 10^{-6}) = 75\text{ nA}$$

This is the AC component of the drain currents. When  $i_{d1} = 75\text{ nA}$ , then  $i_{d2} = -75\text{ nA}$ . The overall drain currents are written as

$$i_{D1} = 20\text{ }\mu\text{A} + (75\text{ nA}) \cdot \sin(2\pi 1\text{kHz} \cdot t) \text{ and } i_{D2} = 20\text{ }\mu\text{A} - (75\text{ nA}) \cdot \sin(2\pi 1\text{kHz} \cdot t)$$

The simulation results are seen in Fig. 22.10. ■



**Figure 22.10** Simulation results for Ex. 22.4.

### AC Gain with a Current Mirror Load

To determine the AC gain of the diff-amp with a current mirror load, Fig. 22.6, consider the small signal model seen in Fig. 22.11. We've replaced the diode-connected MOSFET, M3, with a  $1/g_{m3}$  resistor. Also the resistance looking into the drain of M4 ( $r_{o4}$ ) and the resistance looking into the drain of M2 ( $r_{o2}$ ) are drawn explicitly. We assumed that the small-signal resistance looking into the drain of M2 is simply  $r_{o2}$  (for all practical design cases this is a good assumption). Note how, because of the current mirror action, the AC current flowing in M4 is  $i_{d1}$ . The current in M3 is mirrored by M4. The output voltage can be written as

$$v_{out} = (i_{d1} - i_{d2}) \cdot (r_{o2} \parallel r_{o4}) \quad (22.20)$$

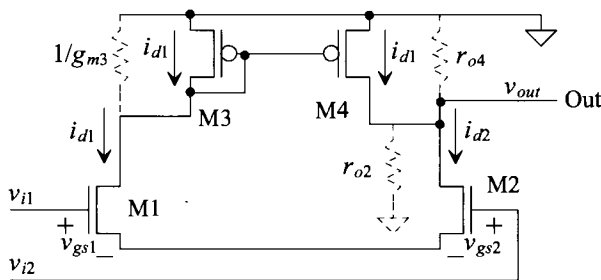
Knowing  $i_{d1} = -i_{d2} = i_d$

$$v_{out} = 2i_d \cdot (r_{o2} \parallel r_{o4}) \quad (22.21)$$

The differential mode gain is then written with the help of Eq. (22.19) as

$$A_d = \frac{v_{out}}{v_{di}} = \frac{v_{out}}{v_{i1} - v_{i2}} = g_m \cdot (r_{o2} \parallel r_{o4}) \quad (22.22)$$

Notice that as the voltage on the gate of M1 increases, the current in M1 (and M3/M4) increases. This causes the output voltage to increase. When the gate potential of M2 increases, so does its drain current, causing the output voltage to decrease. Sometimes the gate potential of M1 is called the *noninverting* input and the gate potential of M2 is called the *inverting* input.



**Figure 22.11** AC circuit of the diff-amp with a current-source load.

**Example 22.5**

Determine the output voltage for the diff-amp seen in Fig. 22.12. Verify your hand calculations using SPICE.

The tail current of the diff-amp conducts  $20\text{ }\mu\text{A}$  so M1 and M2, when the inputs are equal, each conduct  $10\text{ }\mu\text{A}$ . Using the parameters from Table 9.2 and Eq. (22.22), the gain of the diff-amp is

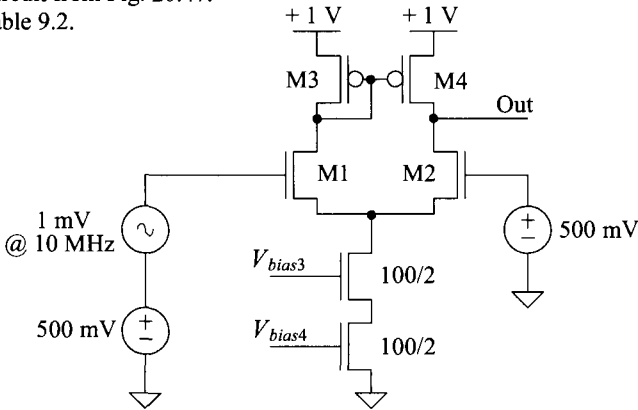
$$A_d = g_m \cdot (r_{on} \parallel r_{op}) = (150 \times 10^{-6}) \cdot \left( \frac{167 \cdot 333}{167 + 333} \times 10^3 \right) = 16.7\text{ V/V}$$

This means that the  $1\text{ mV}$  AC input will appear as a  $16.7\text{ mV}$  AC output. The DC level on the output is  $V_{DD} - V_{SG} = 650\text{ mV}$ . The output voltage is then

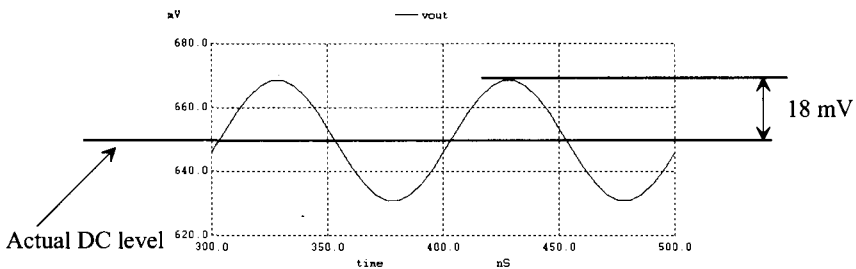
$$v_{out}(t) = 0.65 + (16.7\text{ mV}) \cdot \sin(2\pi 10\text{ MHz} \cdot t)$$

The simulation results are seen in Fig. 22.13. ■

Bias circuit from Fig. 20.47.  
See Table 9.2.



**Figure 22.12** Circuit used in Ex. 22.5.



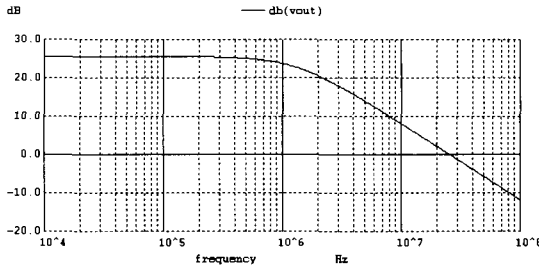
**Figure 22.13** The output of the circuit in Fig. 22.12. Note that the transient analysis doesn't show the beginning of the simulation where the reference circuit is starting up.

**Example 22.6**

Estimate the  $f_{3dB}$  of the diff-amp in Ex. 22.5 if it drives a load capacitance of 1 pF. Verify your hand calculations with SPICE.

The resistance on the output node is  $r_{o2} || r_{o4} = 111 \text{ k}\Omega$ ; therefore, since this is the *high-impedance node*, we can estimate the circuit's 3-dB frequency in the circuit as

$$f_{3dB} = \frac{1}{2\pi \cdot 111 \text{ k} \cdot 1 \text{ pF}} = 1.4 \text{ MHz}$$



**Figure 22.14** The AC response of the diff-amp in Fig. 22.12 when driving a 1 pF load capacitance.

The simulation results are seen in Fig. 22.14. ■

### 22.1.3 Common-Mode Rejection Ratio

An important aspect of the differential amplifier is its ability to reject a common signal applied to both inputs. Often, in analog systems, signals are transmitted differentially, and the ability of an amplifier to reject coupled noise into each line is very desirable. Consider the amplifier shown in Fig. 22.15. The bias current at the sources of M1/M2 has been replaced with its small-signal output resistance. The common-mode signal (the “noise” labeled in the figure) on both inputs is, ideally, rejected by the diff-amp. The diff-amp’s output voltage doesn’t vary. The noise causes the source potentials of M1/M2 to vary (the voltage across the tail current source).

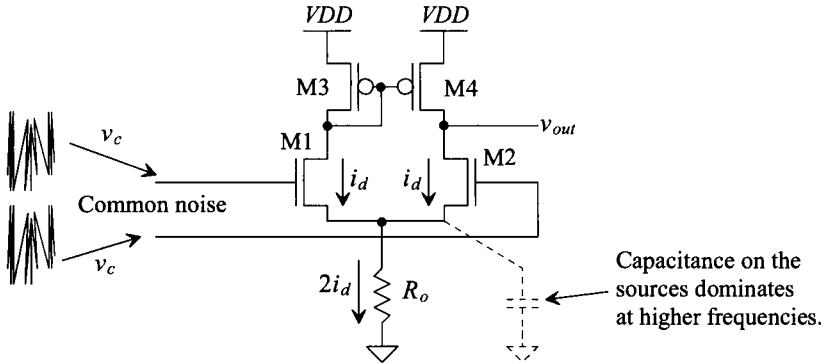
If we apply an AC signal,  $v_c$ , to the gates of M1/M2, equivalent to saying that a common signal is applied to the input of the differential amplifier, we can calculate the common-mode gain. We begin by writing the AC small-signal, common-mode input voltage,  $v_c$ , as

$$v_c = v_{gs1,2} + 2i_d R_o \quad (22.23)$$

The MOSFETs M1 and M2 each source a current,  $i_d$ , through the output resistance of the current source,  $R_o$  (hence the factor of two in this equation). This equation may be rewritten as

$$v_c = i_d \left( \frac{1}{g_m} + 2R_o \right) \approx i_d 2R_o \quad (22.24)$$

The output voltage, because of the symmetry of the circuit, is given by



**Figure 22.15** Calculating the  $CMRR$  of a diff-amp.

$$v_{out} = -i_d \cdot \frac{1}{g_{m3}} = -i_d \cdot \frac{1}{g_{m4}} \quad (22.25)$$

The common-mode gain is then

$$A_c = \frac{v_{out}}{v_c} = \frac{-1/g_{m3,4}}{2R_o} = -\frac{1}{2g_{m3,4}R_o} \quad (22.26)$$

Notice that by increasing the output resistance of the biasing current source, the common-mode gain drops. (This is good because we don't want any common signal to be amplified by the diff-amp.) The common-mode rejection ratio ( $CMRR$ ), in dB, of a diff-amp with a current mirror load is given by

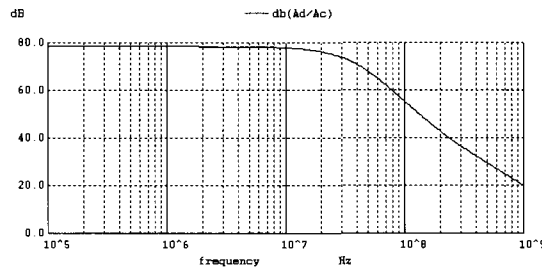
$$CMRR = 20 \cdot \log \left| \frac{A_d}{A_c} \right| = 20 \cdot \log [g_{m1,2}(r_{o2} || r_{o4}) \cdot 2g_{m3,4}R_o] \quad (22.27)$$

The larger the  $CMRR$ , the better the performance of the diff-amp. For the diff-amp in Fig. 22.12, Eq. (22.27) evaluates to 86 dB. As the frequency of the input common-mode signal,  $v_c$ , increases, the  $CMRR$  goes down. This is the result of the impedance of the capacitance to ground on the sources of the diff-pair becoming much smaller than  $R_o$ .

### Example 22.7

Simulate the  $CMRR$  of the diff-amp in Fig. 22.12. Show that the  $CMRR$  falls with increasing frequency.

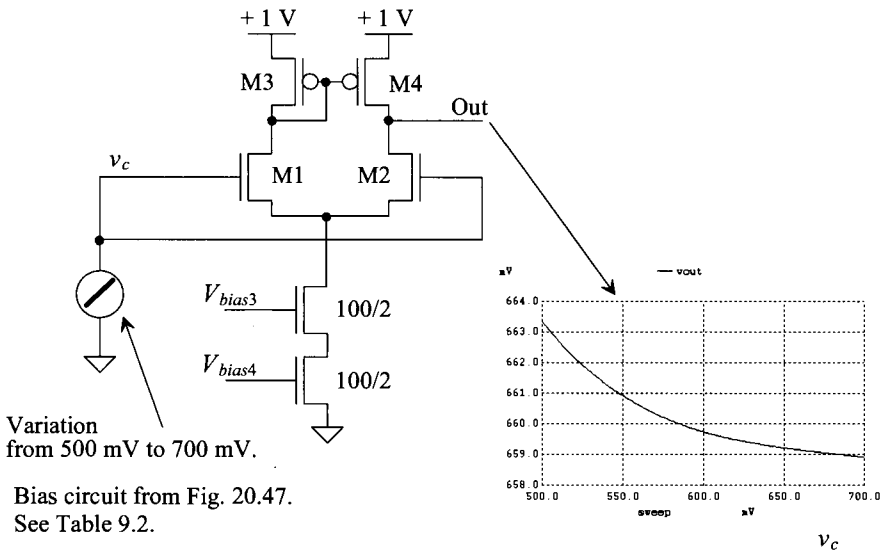
As mentioned above, the (low-frequency) calculated  $CMRR$  is 86 dB. To simulate the  $CMRR$ , we perform AC simulations on two of the diff-amps in Fig. 22.12 so we can simultaneously determine  $A_d$  and  $A_c$ . The ratio of these gains is the  $CMRR$ . Figure 22.16 shows the simulation results. The common-mode gain,  $A_c$ , depends on the DC common-mode voltage. The results seen in Fig. 22.16 are with a common-mode (DC) voltage of 700 mV. Dropping the common-mode voltage down to 500 mV causes the low-frequency  $CMRR$  to drop down to 50 dB. The reason for this is that, with higher DC common-mode voltages, the voltage across the tail current source is higher, so its output resistance is larger. ■



**Figure 22.16** The simulated CMRR for the diff-amp in Fig. 22.12.

*Input-Referred Offset from Finite CMRR*

It's important we understand, in Eq. (22.26), why large  $R_o$  reduces  $A_c$ . Looking at Fig. 22.15, notice that as  $v_c$  varies so too will the voltage across the biasing current source (represented in this figure as  $R_o$ ). As seen in Ch. 20, Fig. 20.4, variations in the voltage across a current source changes the output of the current source (unless its output resistance is infinite). This variation in current causes the current in M1 and M2 to change with variations in  $v_c$ . Changes in the drain current of M1/M2 (and thus M3/M4) causes the output voltage to change. Figure 22.17 shows how the output voltage changes as a function of the DC common-mode voltage. Understanding this circuit is *very important*. If we wanted to hold the diff-amp output voltage constant while varying the common mode voltage, we would have to apply a small voltage difference between the gates of M1 and M2 (*an offset voltage*). We'll revisit this important topic when we discuss op-amp CMRR.



**Figure 22.17** How variation in common-mode voltage affects the output voltage.

22.1.4 Matching Considerations

The fact that performance depends on the matching of devices is another important aspect of differential amplifiers. Layout techniques can be used to minimize the first-order effects of mismatch due to oxide gradients and other process variations. Figure 22.18 shows how the diff-pair would be laid out in a *common-centroid* configuration (see Ch. 5). Common-centroid layout, as the name implies, constructs two devices symmetrically about a common center in the layout. This allows the two devices to cancel process gradients in both the x and y directions and exposes both devices to heat sources in an identical manner. Note how we've attempted to match the parasitic capacitances of the pair (on the gates and drains) and how we've minimized the capacitance at the sources (to keep the *CMRR* high at higher frequencies).

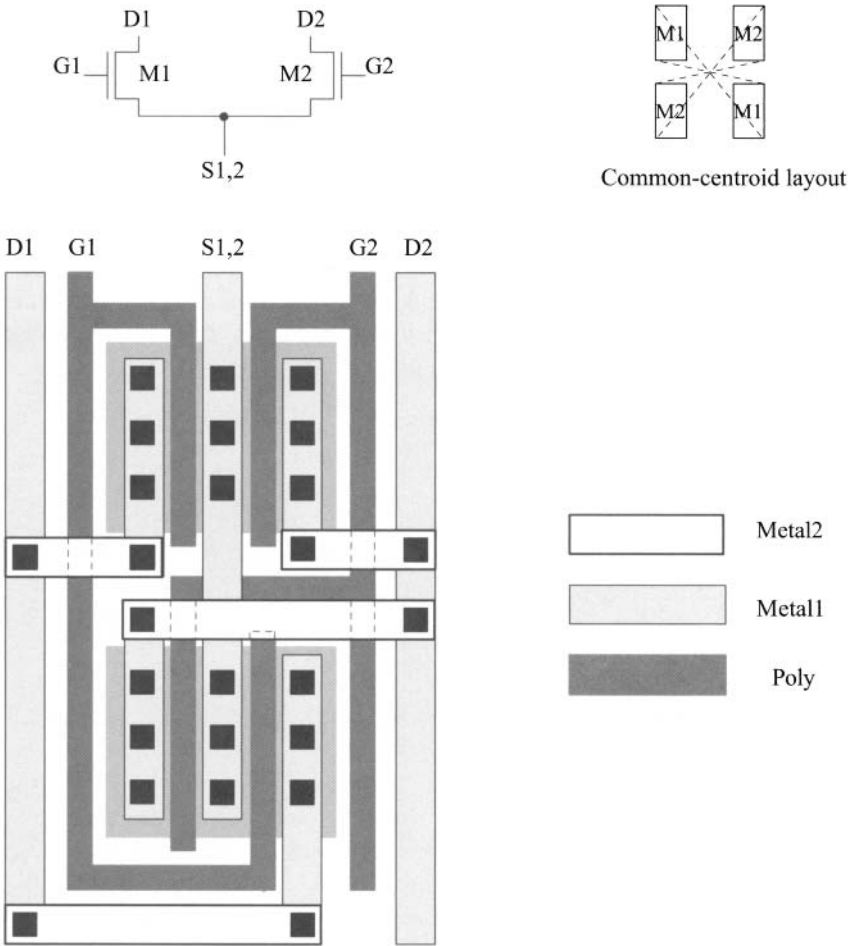


Figure 22.18 Common-centroid layout of a diff-pair.

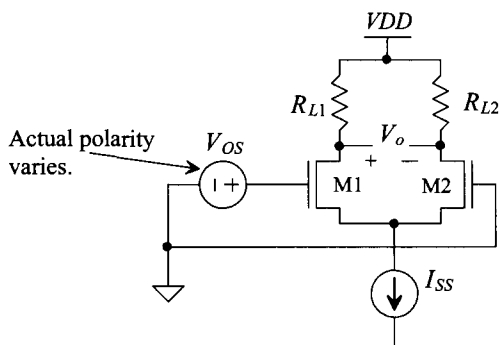
The input-referred offset voltage of a diff-pair, resulting from mismatches in threshold voltage, geometries, and load resistances, can be determined with the help of Fig. 22.19. When the diff-pair is offset-free, connecting the diff-amp's inputs together causes the drain currents and drain voltages of M1 and M2 to be equal ( $V_o = 0$ ). When the load resistors or M1 and M2 are mismatched,  $V_o$  is not zero. (We have mismatch and thus  $V_o$  is offset from its ideal value.). To drive  $V_o$  to zero, we apply a voltage difference between the gates of M1 and M2 (an input-referred offset).

$$V_{OS} = V_{GS1} - V_{GS2} = V_{THN1} + \sqrt{\frac{2I_{D1}}{\beta_1}} - V_{THN2} - \sqrt{\frac{2I_{D2}}{\beta_2}} \quad (22.28)$$

We can define the differences and averages of the threshold voltage, load resistance, and geometry as  $\Delta V_{THN}$  ( $V_{THN1} - V_{THN2}$ ),  $V_{THN}$  (average of  $V_{THN1}$  and  $V_{THN2}$ ),  $\Delta R_L$  ( $R_{L1} - R_{L2}$ ),  $R_L$  (average of  $R_{L1}$  and  $R_{L2}$ ),  $\Delta(W/L)$  [ $(W_1/L_1) - (W_2/L_2)$ ] and  $(W/L)$  [average of  $(W_1/L_1)$  and  $(W_2/L_2)$ ]. Thus, we can make the appropriate substitutions into Eq. (22.28), requiring  $I_{D1}R_{L1} = I_{D2}R_{L2}$ , with the result:

$$V_{OS} = \Delta V_{THN} + \frac{V_{GS} - V_{THN}}{2} \cdot \left[ \frac{-\Delta R_L}{R_L} - \frac{\Delta(W/L)}{(W/L)} \right] \quad (22.29)$$

The threshold voltage mismatch must be reduced using layout techniques. Mismatches resulting from unequal geometry and differences in the load resistance can be reduced by designing with a small  $V_{GS}$  ( $V_{GS}$  close to  $V_{THN}$ ). This should be compared with Eq. (20.8), which shows that using a small  $V_{GS}$  in a current mirror results in a large mirrored current difference.



**Figure 22.19** Determining diff-amp input offset voltage.

### *Input-Referred Offset with a Current Mirror Load*

Suppose the diff-amp with current mirror load, Fig. 22.12, has an offset (the devices aren't perfectly matched). What this means is that the drain potential of M4 (M2) is not equal to the drain potential of M3 (M1 and most likely the drain currents in M1 and M2 aren't equal as well). The input-referred offset would then be the voltage difference on the gates of M1 and M2 needed to drive the output voltage to the correct value. Using Eq. (22.22), we can write the input-referred offset in terms of the difference-mode gain as

$$|V_{OS}| = \frac{V_{o,ideal}}{|A_d|} = \frac{V_{o,ideal}}{g_m \cdot (r_{o2} || r_{o4})} \quad (22.30)$$

which indicates that to reduce the input-referred offset voltage we simply need to design the diff-amp with large gain. This is the same result as that of Eq. (22.29), if we pause and think about it. To reduce the offset, we said we design with a small  $V_{GS}$ . If we hold the diff-amp's bias current constant and reduce  $V_{GS}$  (by increasing the widths of M1 and M2), we are increasing  $g_m$ .

Finally, although not clearly indicated by Eq. (22.30) as it is in Eq. (22.29), a mismatch in the diff-pair's threshold voltage can only be reduced by layout techniques (e.g., using common-centroid layout). Using large devices (larger-width MOSFETs), reduces variation in device characteristics and leads to better matching.

### 22.1.5 Noise Performance

The diff-amp with noise sources is seen in Fig. 22.20. The noise from M6 feeds equally into M1 and M2 (see Fig. 21.44 for comments about noise contributions from M5) and thus, because of the current mirror action in M3 and M4, ideally doesn't affect the output voltage. Therefore, we will ignore noise from M6 in the following discussion.

The diff-amp's output noise power spectral density (PSD) is given by

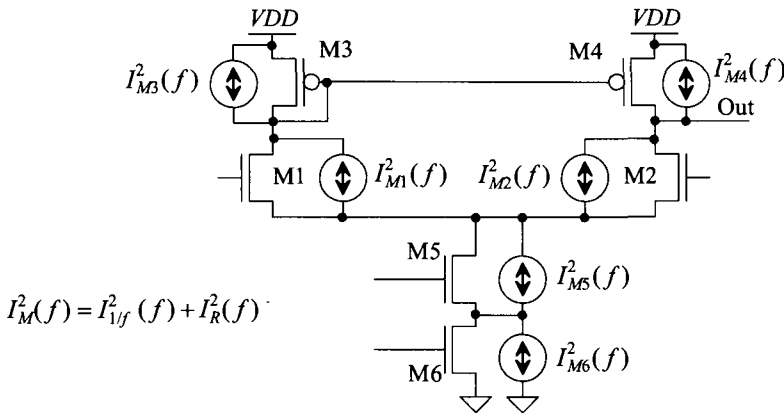
$$V_{noise}^2(f) = (I_{M1}^2(f) + I_{M2}^2(f) + I_{M3}^2(f) + I_{M4}^2(f)) \cdot (r_{o2} || r_{o4})^2 \quad (22.31)$$

The input-referred noise PSD is then

$$V_{noise}^2(f) = \frac{V_{noise}^2(f)}{A_d^2} \quad (22.32)$$

Substituting in the difference-mode gain of the diff-amp, Eq. (22.22), we get

$$V_{noise}^2(f) = \frac{I_{M1}^2(f) + I_{M2}^2(f) + I_{M3}^2(f) + I_{M4}^2(f)}{g_m^2} \quad (22.33)$$



**Figure 22.20** Noise model for the diff-amp.